

Semester Project as Complex Computing Problem

Course Title: Virtual System and Services lab

Instructor: Ms. Bibi Amna

Semester: Spring 2026

Deadline: 22th May 2026

Group Members: **Four** (one in special cases – subject to prior discussion)

A sample scenario is mentioned below. You should create your own scenarios within the scope of the course objectives leveraging the concepts of VMware, Azure, and Docker.

An online E-commerce retail company that operates a Dockerized application for managing its e-commerce platform. The application consists of multiple micro services deployed as Docker containers, including a product catalogue service, user authentication service, and order processing service.

Objective: E-commerce company needs to update its existing Dockerized application to introduce new features and address bugs discovered in the current version.

Deliverables:

- Detailed documentation outlining the updates made to the application
- Updated Docker images for each micro service, tagged with version numbers or unique identifiers.
- Updated deployment configuration files.

PROJECT Gala participation

It is mandatory to participate in project gala. Date/ Time and Venue will be communicated later.

It is mandatory to create a short video (1-2 minutes) along with description of the project.

Marks Distribution

SNo.	Task	Marks
1	Report Writing	40%
2	Functionality/Demo	40%
3	Video on LinkedIn and uploading on GitHub	10%
4	Participation in Project Gala	10%
Total		100%

Timeline

Final Deliverable: Documentation detailing the migration plan and implementation steps, Deadline: 22th May 2026

Project Demos: 22-25th May 2026